ASE 2026 — Neural-driven Computing

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Introduction

Modern large language models and AI toolchains typically run inside heavy software stacks (OS + browser + runtime). Many applications will increasingly embed neural components at many layers of the stack. This project explores whether we can design a far more lightweight computing environment that can directly host perceptron-based neural models as first-class computational elements.

We propose to define a minimal instruction set (ISA) with a small set of neural/tensor primitives, implement an emulator for that ISA, and run small trained networks directly in the emulator. The effort covers ISA design, assembler/loader, model-to-ROM/memory serialization, and verification tooling (including automated black-box tests).

Vision

Create a compact, demonstrable platform where:

A tiny neural model (e.g., a perceptron or small MLP) can be loaded and executed by a minimal runtime / emulator.

The ISA is RISC-like (RISC-V inspired) and extended with a few neural-tensor primitives (matmul, activation, tensor ops).

I/O is minimal: character input (keyboard-like device) and a simple framebuffer output. The pipeline is: Keyboard input → neural model inference → Framebuffer pixel output

The initial demonstration task: train a model that receives short character input and produces a corresponding framebuffer output (e.g., render text or glyphs). The emulator should be capable of loading the model into memory and running inference using the extended ISA.

Use of AI

Use LLM to help with research on the ISA, create AI training data. Also write the emulator for the ISA a bootloader and an assembler. We might also need a “transpiler” to transform the model to something which we can load. Or write our own model creator.